

05506026 Digital Image Processing (Java)

Lab 8

Objectives

1. Understand how to perform erosion and dilation.
2. Understand how to perform other morphological algorithms using erosion and dilation.

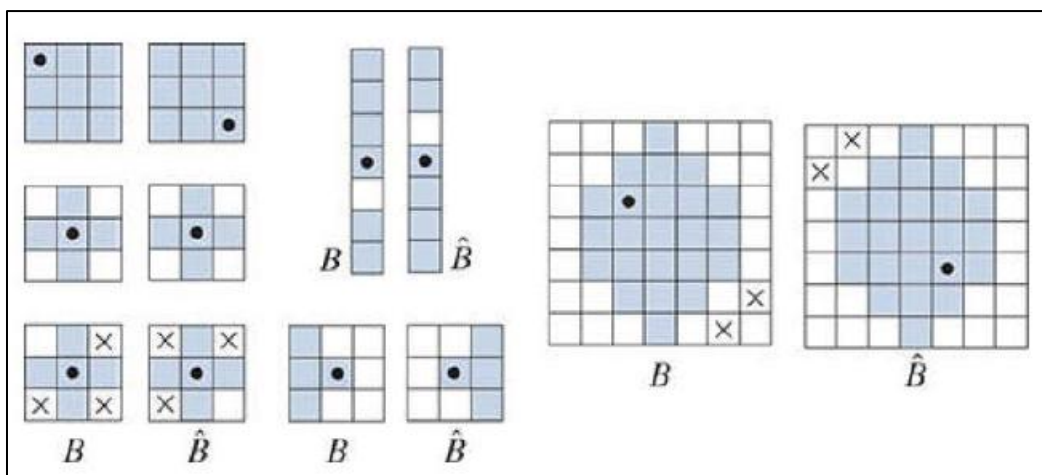
Structuring Elements

Structuring elements is a set of pixels which is used in morphological algorithms. It can consist of foreground and background pixels. Apart from the pixel values, there are also some annotations as follow.

- ● denotes the origin of the set
- X denotes 'don't care' elements

The reflection of an SE B about its origin is $\hat{B}$ defined as

$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$$



We can create structuring elements using code as follow.

```

ImageManager.java
class StructuringElement
{
    public int[][] elements;

    public int width, height;
    public Point origin;

    public ArrayList<Point> ignoreElements;

    public StructuringElement(int width, int height, Point origin)
    {
        this.width = width;
        this.height = height;

        if (origin.x < 0 || origin.x >= width || origin.y < 0 || origin.y >= height)
        {
            this.origin = new Point();
        }
        else
        {
            this.origin = new Point(origin);
        }

        ignoreElements = new ArrayList<>();
        elements = new int[width][height];
    }
}

```

Erosion

Erosion is an operation where If A is a set of foreground pixels (object) and B is structuring elements, the erosion of A by B is defined as

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

Meaning that, the condition of erosion is all foreground pixels in B is a subset of A, then the value at the to the pixel which equivalent to the origin of structuring elements is the foreground value

– white (255) for binary images and $[f \ominus b](x, y) = \min_{(s, t) \in b} \{f(x + s, y + t)\}$ for grayscale images.

In this lab, we will create an erosion method for grayscale images in the future, but we will only do it with binary images for the time being.



The erosion method can be implemented in ImageManager class as follow.

```

ImageManager.java
class ImageManager
{
    ...
    public void erosion(StructuringElement se)
    {
        if (img == null)    return;

        convertToGrayscale();

        BufferedImage tempBuf = new BufferedImage(width, height, img.getType());

        for (int y = 0; y < height; y++)
        {
            for (int x = 0; x < width; x++)
            {
                boolean isEroded = true;
                int min = Integer.MAX_VALUE;

                se_check: for (int i = y - se.origin.y; i < y + se.height -
se.origin.y; i++)
                {
                    for (int j = x - se.origin.x; j < x + se.width - se.origin.x;
j++)
                    {
                        int seCurrentX = j - (x - se.origin.x);
                        int seCurrentY = i - (y - se.origin.y);

```

```

        if (i >= 0 && i < height && j >= 0 && j < width)
        {
            if (!se.ignoreElements.contains(new Point(seCurrentX, se-
CurrentY)))
            {
                int color = img.getRGB(j, i);
                int gray = color & 0xff;

                if (se.elements[seCurrentX][seCurrentY] != gray)
                {
                    isEroded = false;
                    break se_check;
                }
                Else if (min > gray) min = gray;
            }
        }
        else
        {
            isEroded = false;
            break se_check;
        }
    }

    int newGray = 0;

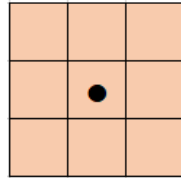
    if (isEroded)
    {
        newGray = min;
    }

    int newColor = (newGray << 16) | (newGray << 8) | newGray;
    tempBuf.setRGB(x, y, newColor);
}

for (int y = 0; y < height; y++)
{
    for (int x = 0; x < width; x++)
    {
        img.setRGB(x, y, tempBuf.getRGB(x, y));
    }
}
}
}

```

In the main class, we must specify the set of structuring elements we are using with their origin and ignored elements. For example, if we want to implement this set of structuring elements.



This will be implemented in the main class.

```
DIPLab8.java
public class DIPLab8
{
    public static void main (String[] args)
    {
        StructuringElement se = new StructuringElement(3, 3, new Point(1, 1));

        for (int i = 0; i < se.height; i++)
        {
            for (int j = 0; j < se.width; j++)
            {
                se.elements[j][i] = 255;
            }
        }
    }
}
```

With this set of structuring elements, the erosion of MandrilB.bmp will be as follow.



Dilation

Dilation is an operation where If A is a set of foreground pixels (object) and B is structuring elements, the dilation of A by B is defined as

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$$

Meaning that, the condition of dilation is at least foreground pixels in $\hat{B}$ is a subset of A, then the value in every pixel which equivalent to the structuring elements is the foreground value – white

$$[f \oplus b](x,y) = \max_{(s,t) \in \hat{b}} \{f(x-s, y-t)\}$$

(255) for binary images and

for grayscale images. Like

erosion, we will create a dilation method for grayscale images in the future, but we will only do it with binary images for the time being.

The dilation method can be implemented as follows.

```

ImageManager.java
class ImageManager
{
    ...
    public void dilation (StructuringElement se)
    {
        if (img == null)    return;

        convertToGrayscale();

        BufferedImage tempBuf = new BufferedImage(width, height, img.getType());

        for (int y = 0; y < height; y++)
        {
            for (int x = 0; x < width; x++)
            {
                boolean isDilated = false;
            }
        }
    }
}

```

```

        se_check: for (int i = y - (se.height - se.origin.y - 1); i < y + se.height -
(se.height - se.origin.y - 1); i++)
        {
            for (int j = x - (se.width - se.origin.x - 1); j < x + se.width -
(se.width - se.origin.x - 1); j++)
            {
                int seCurrentX = se.width - (j - x + se.origin.x) - 1;
                int seCurrentY = se.height - (i - y + se.origin.y) - 1;

                if (i >= 0 && i < height && j >= 0 && j < width)
                {
                    if (!se.ignoreElements.contains(new Point(seCurrentX, seCur-
rentY)))
                    {
                        int color = img.getRGB(j, i);
                        int gray = color & 0xff;

                        if (se.elements[seCurrentX][seCurrentY] == gray)
                        {
                            isDilated = true;
                            break se_check;
                        }
                    }
                }
                else
                {
                    isDilated = false;
                    break se_check;
                }
            }
        }

        if (isDilated)
        {
            int max = Integer.MIN_VALUE;
            for (int i = y - (se.height - se.origin.y - 1); i < y + se.height -
(se.height - se.origin.y - 1); i++)
            {
                for (int j = x - (se.width - se.origin.x - 1); j < x + se.width -
(se.width - se.origin.x - 1); j++)
                {
                    if (i >= 0 && i < height && j >= 0 && j < width)
                    {
                        int color = img.getRGB(j, i);
                        int gray = color & 0xff;

```

```

        if (max < gray) max = gray;
    }
}

int newGray = max;
int newColor = (newGray << 16) | (newGray << 8) | newGray;
tempBuf.setRGB(x, y, newColor);
}
}

for (int y = 0; y < height; y++)
{
    for (int x = 0; x < width; x++)
    {
        img.setRGB(x, y, tempBuf.getRGB(x, y));
    }
}
}
}

```

With the same set of structuring elements as erosion method above, the dilation of MandrilB.bmp will be as follows.

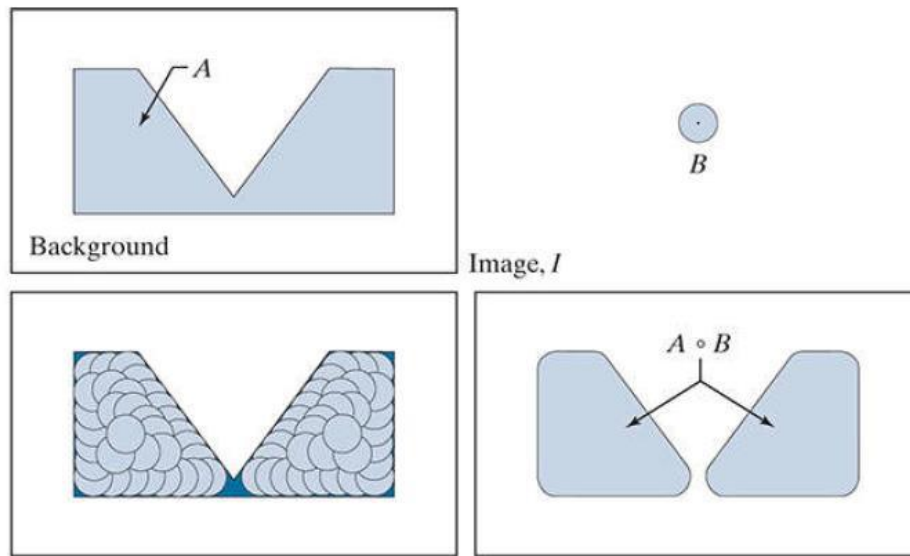


Morphological algorithms

All morphological algorithms are based on erosion and dilation. For example, the opening of set A by structuring element B is defined as

$$A \circ B = (A \ominus B) \oplus B$$

Or erosion then dilation.



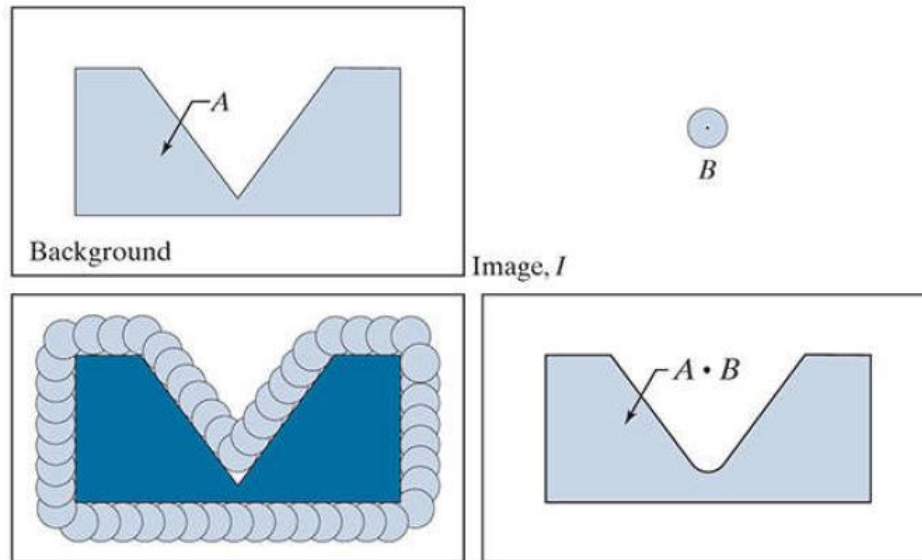
The opening of MandrilB.bmp is as follows.



On the other hand, the closing of set A by structuring element B is defined as

$$A \bullet B = (A \oplus B) \ominus B$$

Or dilation then erosion.



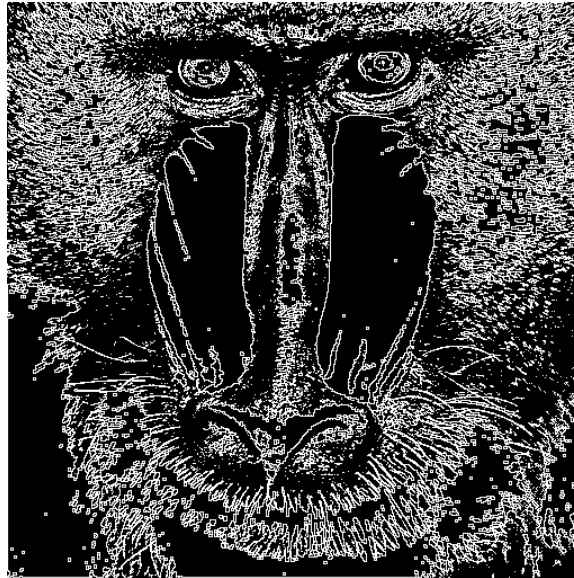
The closing of MandrilB.bmp is as follows.



We can implement more using just erosion and dilation. For example, boundary extraction can be done using structuring elements as above by

$$\beta(A) = A - (A \ominus B)$$

The boundary extraction of MandrilB.bmp is as follows.



Quest 012

Implement the method of boundary extraction.

The player who finishes this task will be awarded **15 EXP** and **2 Moves**.

Deadline: 7 September 2025